Topic 1.4 - Polynomial Functions and Rates of Change

Practice Worksheet

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: ____________________ Class: ______________ Date: ______________

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. In context, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Use finite differences and polynomial structure to identify or justify a likely degree.
- Analyze polynomial rates, turning behavior, zeros, and representations without overclaiming from secant evidence.
- Use technology responsibly to locate features and critique a polynomial model's domain.

Section A: Foundation and representation

Question 1

Finite differences | No calculator

For outputs $2, 4, 18, 56, 130$ at $x = 0, 1, 2, 3, 4$, identify the least degree.

Question 2

Missing value | No calculator

A cubic table has outputs $1, 4, 15, 40, k$ for inputs $0, 1, 2, 3, 4$. Find k if third differences are constant.

Section A continued

Question 3

Structure | No calculator

State maximum possible real zeros and turning points for a degree-6 polynomial. Must both maxima occur?

Question 4

Rates | No calculator

For $P(x) = x^3 - 4x$, find average rates on $[-2, 0], [0, 2], [2, 3]$.

Question 5

Local estimate | Calculator permitted

Given $F(2.995) = 17.011, F(3.005) = 16.989$, estimate the rate at 3.

Section B: Application and reasoning

Question 6

Rate pattern | No calculator

One-unit rates over four intervals are $-5, 1, 13, 31$. Identify a likely degree of the original polynomial.

Question 7

Parameter | No calculator

For $P = x^3 + kx^2 - 4x + 2$, the rate on $[0, 2]$ is 6. Find k.

Question 8

Construction | No calculator

Construct a cubic with zeros $-1, 2, 5$ and $P(0) = 20$.

Section C: Challenge and error analysis

Question 9

Connections | No calculator

Two polynomials share leading term $3x^5$. State one feature they must share and two they need not.

Question 10

Error analysis | No calculator

A cubic appears to have one zero in a calculator window. Explain how other zeros could be hidden and how to verify.

Question 11

Model critique | No calculator

A quartic regression fits five measured points exactly. Give two reasons this is not enough for a ten-year forecast.

Question 12

Challenge | No calculator

Construct a degree-4 polynomial with exactly two distinct real zeros, each of multiplicity 2, and describe its graph at those zeros.

Section D: AP-style multiple choice

MCQ 1

No calculator

Which pattern most strongly indicates a cubic for exact equal-step data?

A. Constant first differences

B. Constant second differences

C. Constant third differences

D. Alternating first differences

MCQ 2

No calculator

Maximum possible turning points of degree 8?

A. 7

B. 8

C. 9

D. 16

MCQ 3

No calculator

A polynomial has equal values at $x = 1, 5$. What must be true?

A. It is constant on $[1, 5]$.

B. Its average rate on $[1, 5]$ is 0.

C. It has a zero there.

D. It is quadratic.

MCQ 4

No calculator

A quartic is fitted on $0 \leqslant t \leqslant 6$. Which use is least justified without more evidence?

A. Evaluate at 3

B. Rate on $[2, 4]$

C. Describe a fitted turn in the interval

D. Predict at 60

AP-style free response 1

Calculator permitted - 6 points

On $0 \leqslant t \leqslant 10$, $H(t) = -0.04t^4 + 0.8t^3 - 4.5t^2 + 6t + 28$.

A. Find local extrema to three decimals.

B. Determine absolute extrema on the domain.

C. Calculate the average rate on $[2, 8]$.

D. Give one reason not to use the model for all $t > 10$.

AP-style free response 2

No calculator - 6 points

Let $P(x) = (x+2)(x-1)^2(x-4)$.

A. State degree, leading coefficient, and end behavior.

B. List zeros, multiplicities, and cross/touch behavior.

C. Determine sign on intervals formed by zeros.

D. Find the rate on $[0, 2]$ and interpret equal endpoints.